



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
Continuous Assessment Test – I

Course Name & Code:	Data Structure and Algorithms	CSE 2003
Class Number:	Slot:G2	Exam Duration: 90 Min
		Maximum Marks:50

Answer All Questions

1 **a)** Arrange the following in the increasing order of complexity and plot its growth rate of various function with an example.

i) $n \log n$ ii) n^2 iii) $\log n$ iv) $n^3 \log n$ v) $n (\log n)^2$ vi) n (5) marks

b) Explain the need of analyzing an algorithm. (5) marks

2 **a)** Prove the following given function

$f(n) = 10n^3 + 3n^2 + 6,$ P.T $f(n) = O(n^3)$ (5) marks

b) Design the data structures that you would implement to efficiently realize “search a number in a given one dimensional array”. Discuss its worst-case complexity in computation with varying search key values. (5) marks

3 **a)** Explain circular queue with an example. (5) marks

b) Find the time complexity of recursive factorial given the

$$\begin{cases} T(n) = a, n \leq 1 \\ T(n) = T(n-1) + b, n > 1 \text{ for some constants } a \text{ and } b \end{cases} \quad (5) \text{ marks}$$

4 a) Convert the corresponding infix expression into postfix form using reverse polish with appropriate intermediate steps.

$$(a*b)+(r*c)+w$$

5 marks

b) It is sometimes necessary to perform stack operation with variable length strings. Write an algorithm for push and pop procedures that manipulate such strings stored on a stack consisting of a fixed length character array.

5 marks

5 a) Write a note on Master theorem and explain where master theorem cannot be applied.

5marks

b) Solve $T(n) = 6T\left(\frac{n}{2}\right) + O(n)$ using master method

5 marks